

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 - Digital Electronics
Spring Term 2026

Lab. 2- Transistor Parameter Extraction

Issued: 1/28/2026

Due: 1/30/2026

Question 1 (12 points):

Objectives:

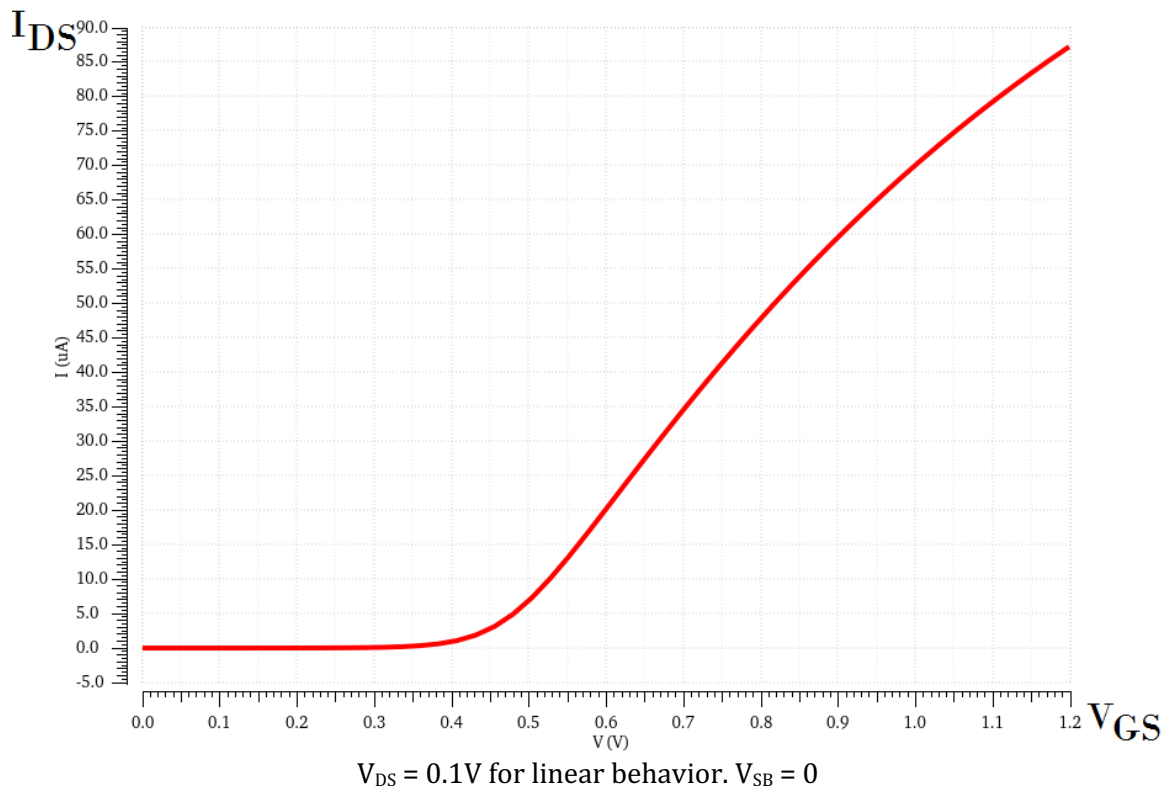
- From I_{DS} - V_{GS} and I_{DS} - V_{DS} curves determine key MOSFET parameters.
- Learn impact of short channel behaviors on device performance.

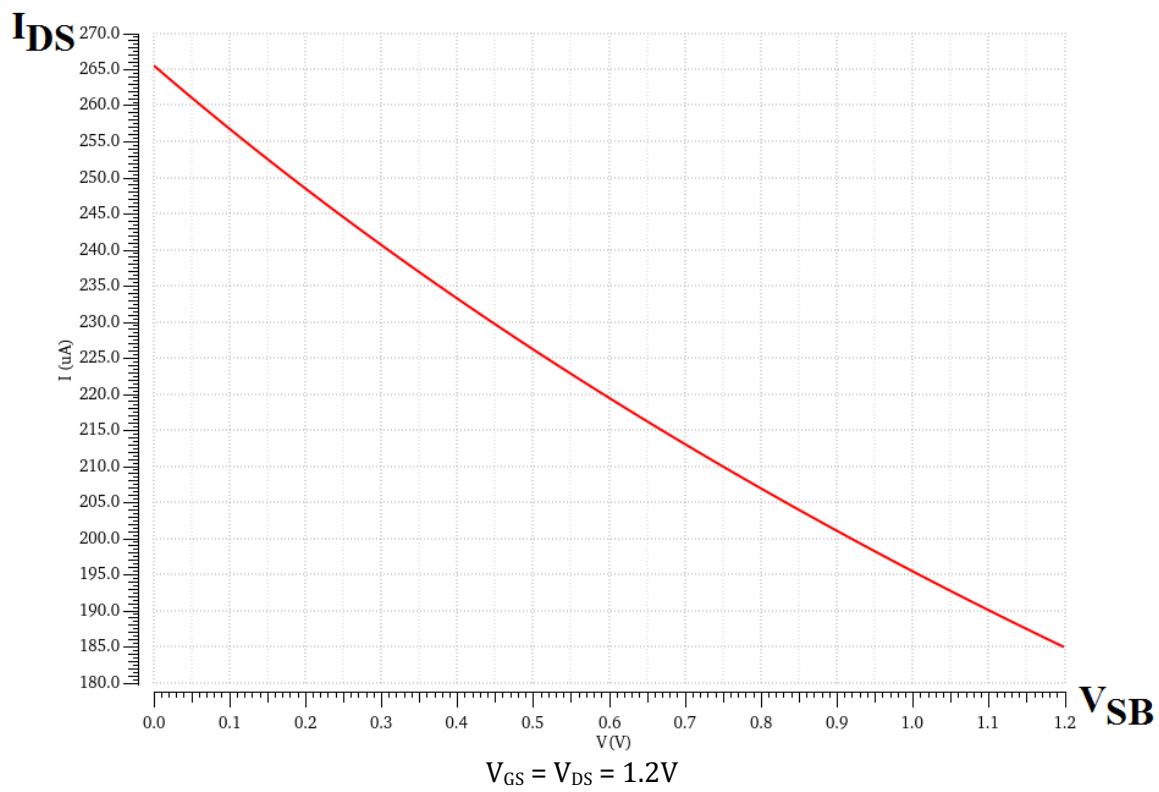
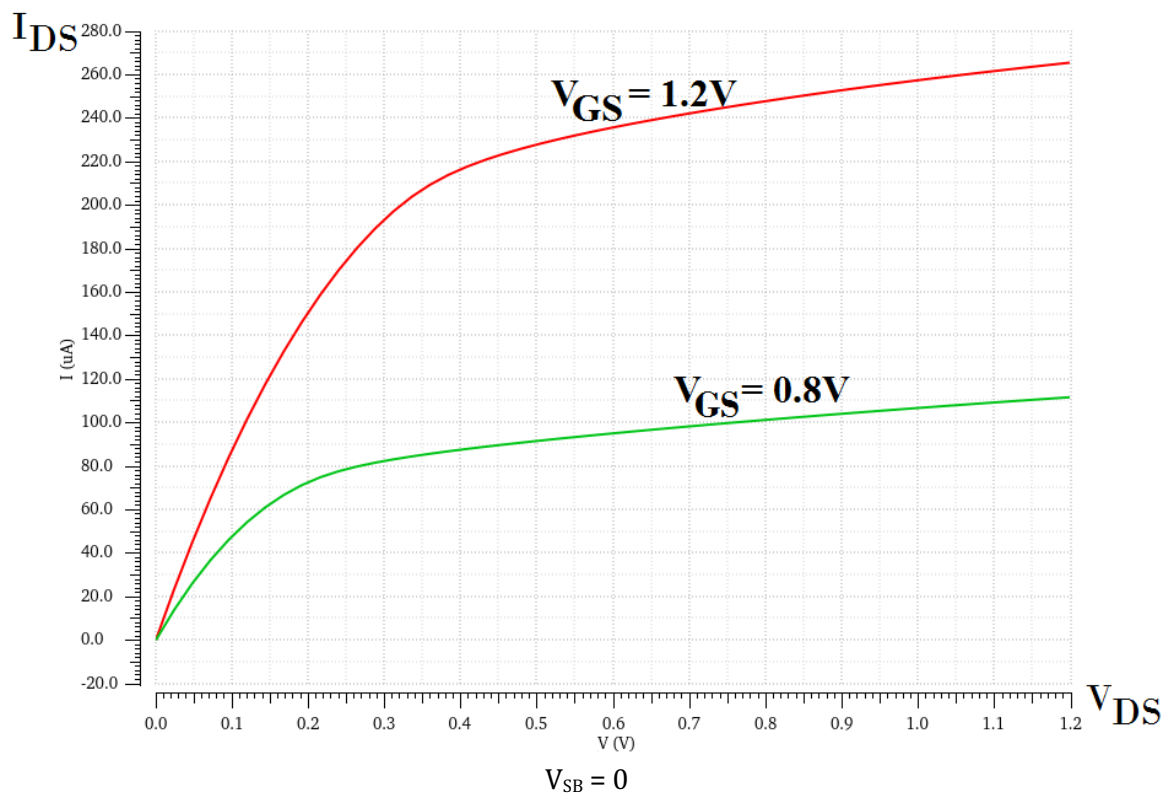
Given

The following plots are simulations of a 130nm NMOS using Cadence Spectre, equivalent to a SPICE level-54 simulation. Simulation parameters are tailored based upon fabricated results for highly accurate simulation plots.

All simulations are with the same transistor with the following characteristics:

- $W/L = 5$
- $L = 120 \text{ nm}$
- $t_{OX} = 3.2 \text{ nm}$ ($C_{OX} = 1.1 \cdot 10^{-6} \text{ F/cm}^2$)
- $2\phi_F = 0.88 \text{ V}$





There are three models of NMOS behavior that you will walk through in this question:

1. Ideal Long channel NMOS:

$$I_{DS(sat)} = \frac{1}{2} \mu_n C_{OX} \frac{W}{L} (V_{GS} - V_T)^2$$

$$I_{DS(lin)} = \mu_n C_{OX} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

2. NMOS with channel length modulation:

$$I_{DS(sat)} = \frac{1}{2} \mu_n C_{OX} \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

$$I_{DS(lin)} = \mu_n C_{OX} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right] (1 + \lambda V_{DS})$$

3. NMOS with channel length modulation and horizontal critical field limitation (E_c):

$$I_{DS(sat)} = W v_{sat} C_{OX} \frac{(V_{GS} - V_T)^2}{(V_{GS} - V_T + E_c L)} (1 + \lambda V_{DS})$$

$$I_{DS(lin)} = \frac{\mu_n C_{OX}}{\left(1 + \frac{V_{DS}}{E_c L}\right)} * \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right] (1 + \lambda V_{DS})$$

Perform the following tasks. Calculate the following values using the plots above. Whenever applicable always state plots and data points used.

- a) State the threshold voltage, V_{T0}
 - i. V_{T0} can either be calculated or determined by observation.
- b) Calculate the channel length modulation parameter, λ
- c) Calculate the body effect coefficient, γ
 - i. Use the method of finding ΔV_T between two points and solving for γ in the following equation:
$$V_T = V_{T0} + \gamma (\sqrt{V_{SB} + |2\phi_F|} - \sqrt{|2\phi_F|})$$
- d) Calculate the electron mobility, μ_n
 - i. Include the impact of λ in all calculations for part d.
 - ii. With high channel doping μ_n is expected to be between 350 and 500 cm²/Vs. How does your value compare? What does this mean?
 - iii. Calculate I_{DS} with a μ_n of 350 cm²/Vs, $V_{GS}=V_{DS}=1.2V$, $V_{SB}=0$.
- e) Determine $V_{DS(sat)}$, the V_{DS} at the transition point between linear and saturated operation in the following scenarios when $V_{GS} = 1.2V$, $V_{SB}=0$:
 - i. What is the expected $V_{DS(sat)}$ if this were a long-channel MOSFET?
 - ii. What is the observed $V_{DS(sat)}$?
- f) Based upon the observed $V_{DS(sat)}$, calculate $E_c * L$
 - i. With this $E_c * L$ value, λ , and other extracted values calculate I_{DS} when $V_{GS}=V_{DS}=1.2V$, $V_{SB}=0$.
 - a) How does this compare to the plotted result?
 - b) How does this compare to the I_{DS} result in part 1.d.iii?

Question 2 (8 points): Based on Problem 3.5(D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

1. Observe the effect of temperature on desirable $I_{DS(\text{linear})}$ and $I_{DS(\text{sat})}$ currents, as well as undesirable $I_{DS(\text{subthreshold})}$ current.
2. Learn how to perform DC-sweep plots in LTspice.

When a transistor is on, we want it to pass as much current as possible, and when it is off to pass as little current as possible to form an ideal switch. Use the provided model library file **130nm_bulk.txt** on LMS to perform LTspice simulations for this section with the following parameters:

- $W/L = 5$
- $L = 120\text{nm}$
- $V_{DS} = 1.2\text{V}$
- $V_{SB} = 0$

Perform the following tasks. You will need to set up an LTspice NMOS simulation schematic with V_{GS} , V_{DS} , and V_{SB} voltage sources with $V_{DS}=1.2\text{V}$ and $V_{SB}=0$. Short 1-2 sentence answers are expected for analysis questions.

- a. Plot the I_{DS} **and the log of I_{DS}** sweeping V_{GS} from 0 to 0.4V to plot current in the subthreshold region at the following temperatures:
 - i. -40°C , 25°C , 125°C
 - ii. Record all screenshots.
 - iii. What is the overall correlation between subthreshold current and temperature? Are higher or lower temperatures desirable? Why?
- b. Plot the I_{DS} sweeping V_{GS} from 0.4 to 1.2V to plot current in the active region at the following temperatures:
 - i. -40°C , 25°C , 125°C
 - ii. Record all screenshots.
 - iii. What is the overall correlation between active region current and temperature? Are higher or lower temperatures desirable? Why?

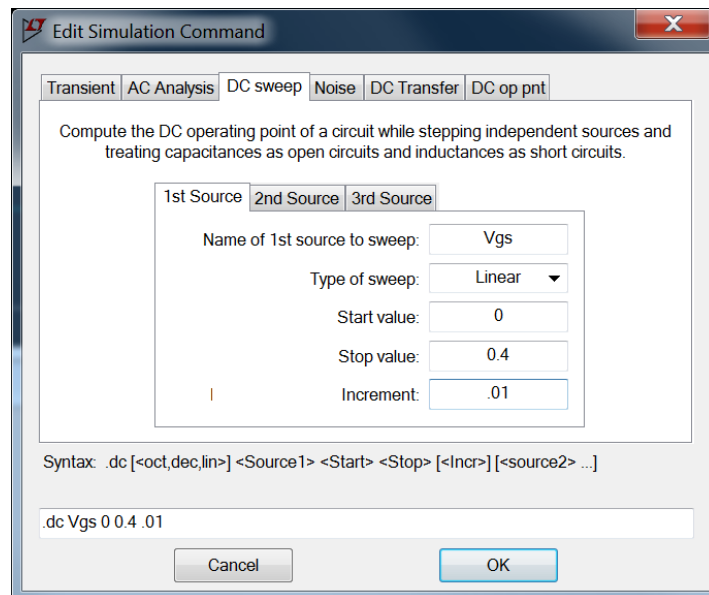
LTspice

SPICE directives:

- For this lab you do not need any **.model** spice directives. The included **130nm_bulk.txt** file defines all needed parameters other than length and width, which are defined in the NMOS symbol.
- To set a simulation temperature, add the **.TEMP** spice directive. For example, to set a 100C temperature:
 - **.TEMP 100**
- For the SPICE BSIM3 Level-49 NMOS model, you must use the **.include** or **.lib** SPICE directive for LTspice to locate the file. Below is an example:
 - **.lib C:\rpi teaching\digital electronics\130nm_bulk.txt**

DC-sweep plot:

- To add a simulation, click the **Run** icon in the LTspice top toolbar.
- Instead of a **DC op pnt** simulation like we did in lab 1, we will now do a **DC sweep** simulation.
 - Match **Name of 1st source to sweep** to the name of whatever is biasing your NMOS V_{GS} .
 - Set **Type of sweep** to **Linear**.
 - Enter the initial and final VGS voltages in **Start value** and **Stop value**.
 - Set **Increment** to something small, say 0.01V.
 - All other fields can be ignored. Click **OK**.



Above is an example of how to set a DC sweep simulation.

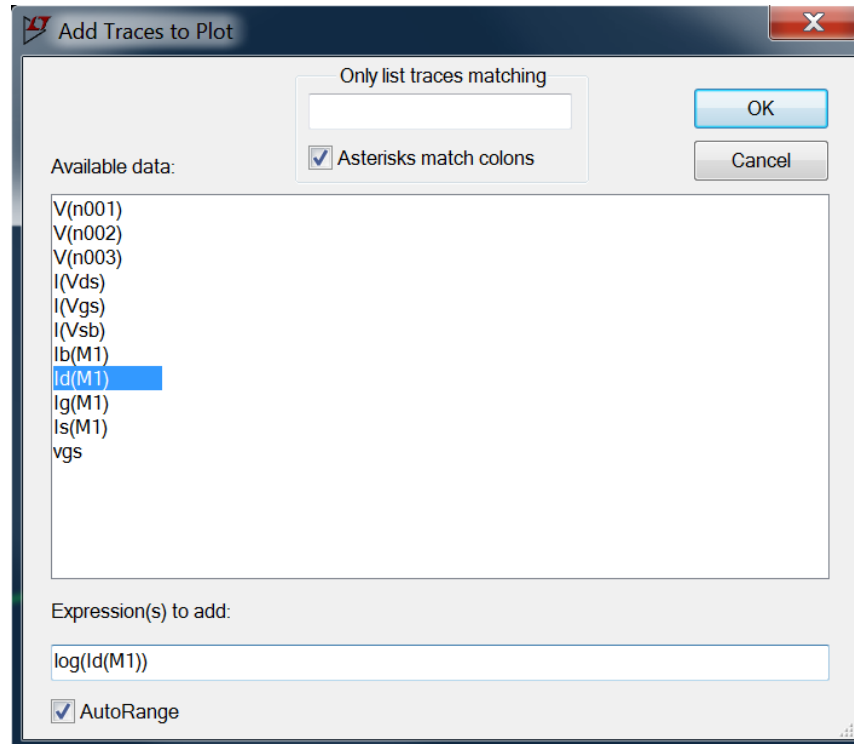
```
.TEMP 25  
.lib C:\rpi teaching\digital electronics\130nm_bulk.txt  
.dc Vgs 0 0.4 .01
```

Your final SPICE directives after adding the DC sweep will look similar to above.

Log scale plot:

To plot on a log scale do the following:

1. Run a simulation
2. **Right-click** the plot background
3. **Add traces**
4. Select what you want to plot, in this case **Id(M1)**
5. Wrap the trace with the **log()** command
6. Click **OK**



Report Submission

Submit your work on LMS under **Labs**

- Show your work.
 - Include images of your hand calculations with work and final results.
 - Include screenshots of LTspice.
 - Include all required discussion.
- Due date: Friday at 11:59pm.